Rates of Chemical Reactions (p162-166)

Fast and Slow Reactions

1. What is the rate of reaction?

2. Give examples of reactions which have a **fast** rate of reaction.

3. Give examples of reactions which have a **slow** rate of reaction.

Controlling the Rate of Chemical Reactions

1. Give an example of a chemical reaction in your body which can be increased or decreased.

2. List five factors which affect the rate of reaction.
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Temperature

1. Increasing the temperature normally increases the rate of reaction. Give two reasons why this occurs.
   i.

   ii.

2. Give two examples of situations where decreasing the temperature decreases the rate of reaction.

Concentration

1. What does the term concentration mean? Use an example as part of your answer.

2. Draw a diagram of a concentrated solution and a dilute solution.

3. Explain why increasing the concentration of reactants increases the rate of reaction.

4. In order to slow down the rate of reaction, the concentration of reactants can be reduced. Describe a situation

Agitation

1. What is another name for agitation?

2. How does agitation increase the rate of reaction?

3. Describe an example where agitation helps speed up a reaction. Use a diagram as part of your answer.

Surface Area of Reactants

1. How can you increase the rate of reaction between hydrochloric acid and calcium carbonate (chalk)?

2. Why does breaking a solid into smaller pieces increase the reaction rate? Use a diagram in your answer.

Catalysts

1. What is a catalyst?

2. How do catalysts help in a reaction?

3. Give an example of how a catalyst is used in everyday life? (hint: more on p166)

Enzymes

1. What are enzymes?

2. Name an enzyme in saliva and describe what it does.